Antifragile Value Scoring System (AFV 2.1): A Quantitative Framework for Robust Stock Selection

Overview

The Antifragile Value (AFV) scoring system is a proprietary, rules-based metric designed to evaluate stocks based on principles of antifragility, as inspired by Nassim Nicholas Taleb's framework. It prioritizes resilience to volatility, convexity in payoffs (limited downside with disproportionate upside), and avoidance of fragility from debt, cyclicality, or overvaluation. The score aggregates multiple financial and qualitative factors into a composite value, with AFV 2.1 introducing refinements for net income stability and margin adjustments.

The system is computed monthly using trailing twelve-month (TTM) and historical data from sources like Yahoo Finance (or Polygon for enhanced accuracy in v2.2 proposals). Stocks are scored on a scale typically ranging from -10 to +10, with positive scores indicating antifragile characteristics suitable for long-term holding in uncertain markets. The framework emphasizes "real-world" utility (e.g., commodities, essentials) over speculative sectors.

Key equation for AFV 2.1 score: $$AFV_{2.1} = RP_{2.1} + S + G + D + T + VD$$ Where:

- ($RP_{2.1}$): Refined Return Potential (cash flow-based, capped at [-3, 5])
- (S): Sector/Industry Score ([-1, 1])
- (G): Geographic Stability Score ([-1, 1])
- (D): Debt Resilience Score ([-1.5, 1])
- (T): Trend Momentum Score ([-1, 1])
- (VD): Valuation-Dividend Score ([-1, 0.5])

Negative overrides (e.g., -1000) are applied for data unavailability or critical fragility (e.g., negative equity).

1. Refined Return Potential (RP and RP 2.1)

RP quantifies the stock's cash generation efficiency relative to market capitalization, penalized for volatility and stress periods. It uses a logistic function to reward high free cash flow (FCF) yields while capping extremes.

FCF Yield: $$FCF_{yield} = \frac{\overline{FCF}_{TTM}}{Market\ Cap}$$ Where ($\overline{FCF}_{TTM} = \overline{OCF}_{TTM} + \overline{CapEx}_{TTM}$) (CapEx is negative), averaged over last 4 quarters, normalized to EUR.

OCF Margin: $$\overline{OCF}_{margin} = \frac{\overline{OCF}_{TTM}}{\overline{Revenue}_{TTM}}, \quad min(OCF_{margin}) = \min(OCF_{margins, last\ 4q})$$ Capped at [-2, 2] to avoid outliers.

OCF Margin Volatility (CV): $$CV = \frac{\sigma(OCF_{margins, last\ 3y+})}{|\mu(OCF_{margins})|}$$ Requires ≥3 years; skips if insufficient.

RP (Base): If ($FCF_{yield} \leq 0$): RP = -2
Else: $$RP = \frac{7}{1 + e^{-30(FCF_{yield} - 0.15)}} - 2$$ Adjustments:

- If ($\overline{OCF}_{margin} < 0$): RP = -2
- Scale by margin level (e.g., <0.05: ×0.7; >0.20: ×1.2)
- If ($min(OCF_{margin}) < 0$): RP = -2
- If ($min(OCF_{margin}) < 0.5 \times \overline{OCF}_{margin}$): ×0.7
- Volatility penalties: CV >1: -2; >0.3: ×0.5; >0.15: ×0.8; <0.05: ×1.1 Clamped to [-3, 5].

RP 2.1 (Enhanced): Incorporates net income checks for profitability consistency. Net Income Check: Flags if any of last 4 annual net incomes <0; computes ($\overline{Net}_{margin}$).

Base: $$RP_{2.1} = \frac{5.5}{1 + e^{-35(FCF_{yield} - 0.15)}} + 0.5$$ Adjustments:

- Negative net income: ×0.5
- OCF margin multiplier: 0.8 + 2×OCF_margin (clamped [0.7, 1.3])
- Min margin deterioration: Gradual ×[0.5, 0.8]
- Volatility: Threshold 0.20; penalties linear [0.5, 0.8] if >threshold; ×1.1 if <0.05
- Net margin: <0.05: ×0.5; >0.15: ×1.1 Clamped [-3, 5].

Financial sector override: RP/RP2.1 = 0 (due to non-comparable cash flows).

2. Sector/Industry Score (S)

Scores based on antifragility: +1 for essentials (e.g., energy, utilities, commodities like oil/copper/mining), 0 for neutral (e.g., chemicals, telecom), -1 for fragile (e.g., tech, airlines, biotech). Industrials use granular industry scores (e.g., +1 for mining/steel, -1 for airlines). Mapped via hardcoded dictionary; defaults to 0.

3. Geographic Stability Score (G)

Scores country of headquarters: +1 for stable regions (e.g., Germany, Finland, US at 0.8), -1 for high-risk (e.g., Russia). Defaults to 0.6. Aims to favor geopolitically robust locations.

4. Debt Resilience Score (D)

Measures balance sheet strength to withstand shocks.

Net Debt: Total Debt - Cash

If Equity > 10% Assets: $$Net\ D/E = \frac{Net\ Debt}{Equity}$$ Scores: <0: +1; <0.5: +0.5; <1.5: 0; <3: -0.5; else -1.5

Fallback if low/negative equity: Net Debt / OCF (similar tiers).

Financials/Industrials overrides: 0 or skipped if fragile sub-industry.

5. Trend Momentum Score (T)

Weighted rate-of-change (ROC) in OCF over last 3 periods (years): $$ROC_w = 0.1 \times ROC_{y-3/y-2} + 0.3 \times ROC_{y-2/y-1} + 0.6 \times ROC_{y-1/y}$$ Scores: ≥0.5: +1; ≥0.2: +0.5; ≥-0.2: 0; ≥-0.5: -0.5; else -1

Penalized if low/negative OCF margin (e.g., <0.1: deduct 0.5-1).

Requires ≥4 years OCF.

6. Valuation-Dividend Score (VD)

Guards against overvaluation:

- P/E >50: -1
- P/E >20: -1 if Div Yield <1%, -0.5 otherwise
- Else if Div Yield ≥3%: +0.5
- Invalid P/E: -0.5

Financials override: 0.

Computation Process

- Fetch data (history, cashflow, financials, balance sheet, info) with 1-month caching.
- Skip recompute if recent score exists.
- Handle errors: Score -1000.
- Monthly DCA: Buy if AFV > threshold (user-defined, e.g., >3).

Limitations & Antifragile Rationale

AFV avoids prediction, focusing on preparation: High scores favor stocks that "gain from disorder" (e.g., commodity producers in inflation/geopolitical scenarios). No market timing; emphasizes subtraction of risks (e.g., high CV, debt). For v2.2, propose adding crowding (volume surge penalty), convexity (low beta bonus), and precious metals boost.

This framework supports slow, macro-drifted investing, as in your barbell strategy (safe T-bills/gold + convex stocks + tail puts).

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